On a conjecture of Ayala–Francis–Rozenblyum

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University of Vienna
(work done in part in Zurich)

1. The smooth realisation problem

2. The stratified generalisation

3. The conjecture and the counterexamples

4. The main theorems

5. The proofs of the counterexamples

6. Outlook: ∞-categories via geometric topology?

The smooth realisation problem

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The stratified generalisation

∞-groupoids $\rightsquigarrow$ ∞-categories

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- (M, s): appropriately *smooth*

Recall:

$$\mathrm{Sing}_n(M) = \{\Delta^n \to M\}$$

where

$$\Delta^n = \{(t_0, \ldots, t_n) \in \mathbb{R}^{n+1} : t_0 + \cdots + t_n = 1\}$$

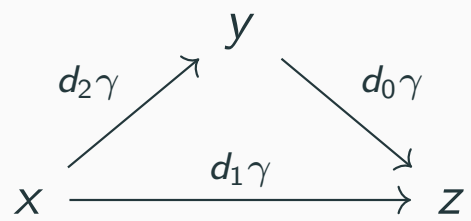
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- Sing_0 : points $x \in M$ are objects,
- Sing_1 : paths $x = \gamma(0) \xrightarrow{\gamma} \gamma(1) = y$ are 1-morphisms
- Sing_2 : homotopies $\gamma\colon \Delta^2 \to M$:



are 2-morphisms '$d_2\gamma \circ d_0\gamma \Rightarrow d_1\gamma$', etc.

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e.g.

$$s\colon \mathbb{R}^2 \to \{0 < 1\},$$
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above: $(\mathbb{R}^2, s) \cong C(S^1)$

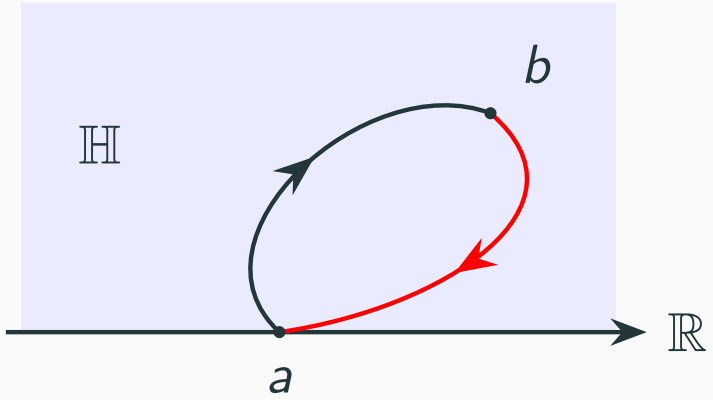
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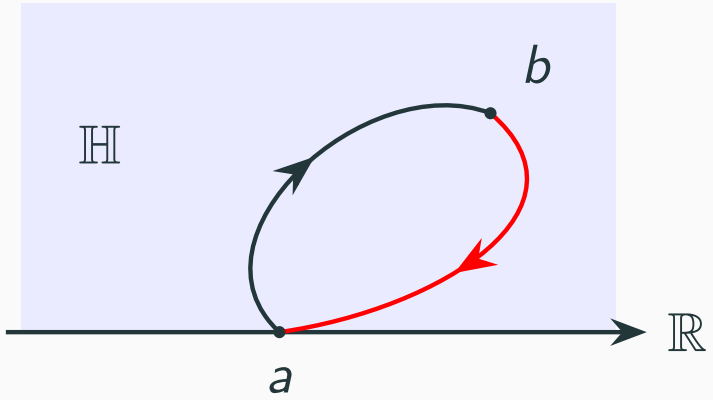
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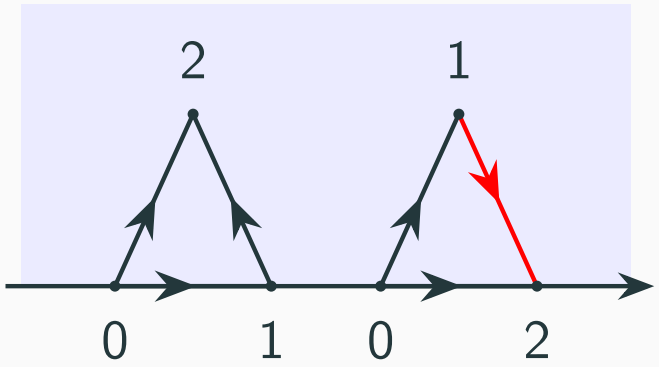


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[Treumann, Lurie]

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- Without non-trivial endomorphisms: $|* \xrightarrow{S^1} *| \simeq \Sigma S^1$

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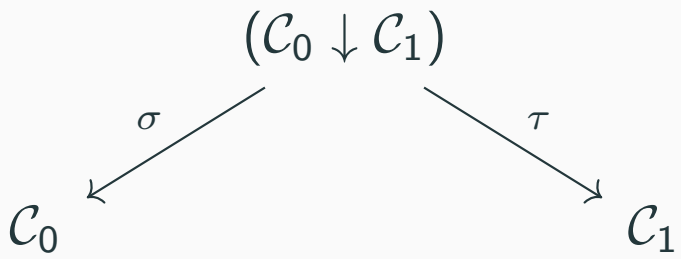
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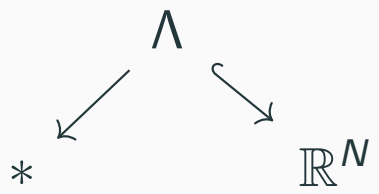


The conjecture and the counterexamples

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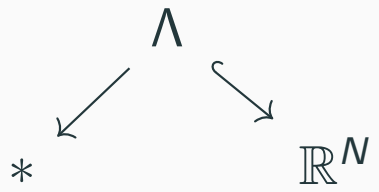
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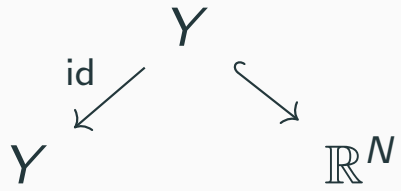
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such that $\mathcal{E} \neq \mathrm{Exit}(M)$ for any compact M, with Y closed + non-homology-sphere.

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Globally: Spherical blow-up:

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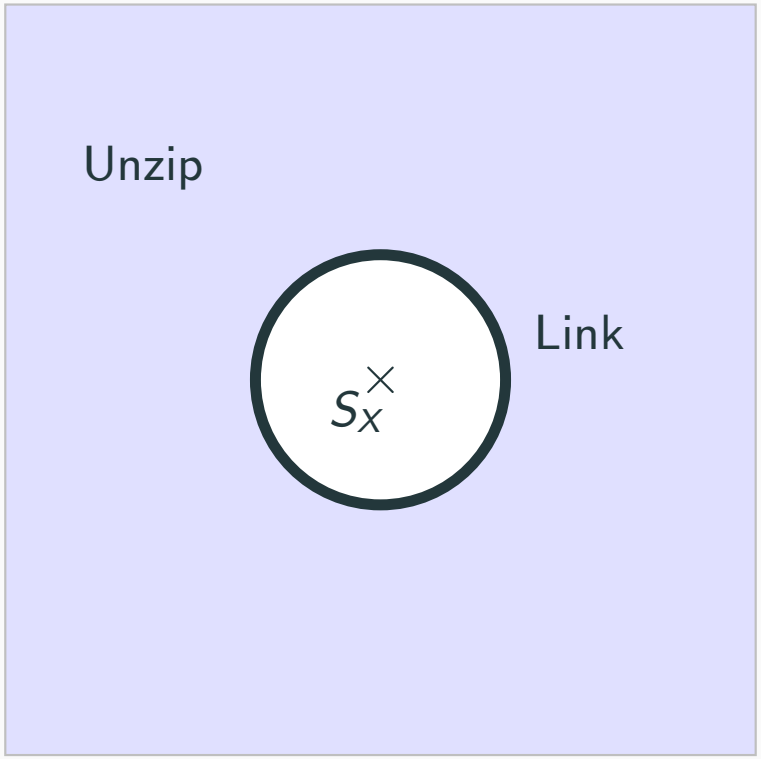
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The cone:

$$\begin{array}{ccc} \{0\} \times Z & \hookrightarrow & [0,\infty) \times Z \\ \downarrow & & \downarrow \\ * & \xhookrightarrow{\text{cone pt}} & C(Z) \end{array}$$

‘Polar coordinates’

$$\mathbb{R}^2 \to \{0 < 1\},$$
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The main theorems

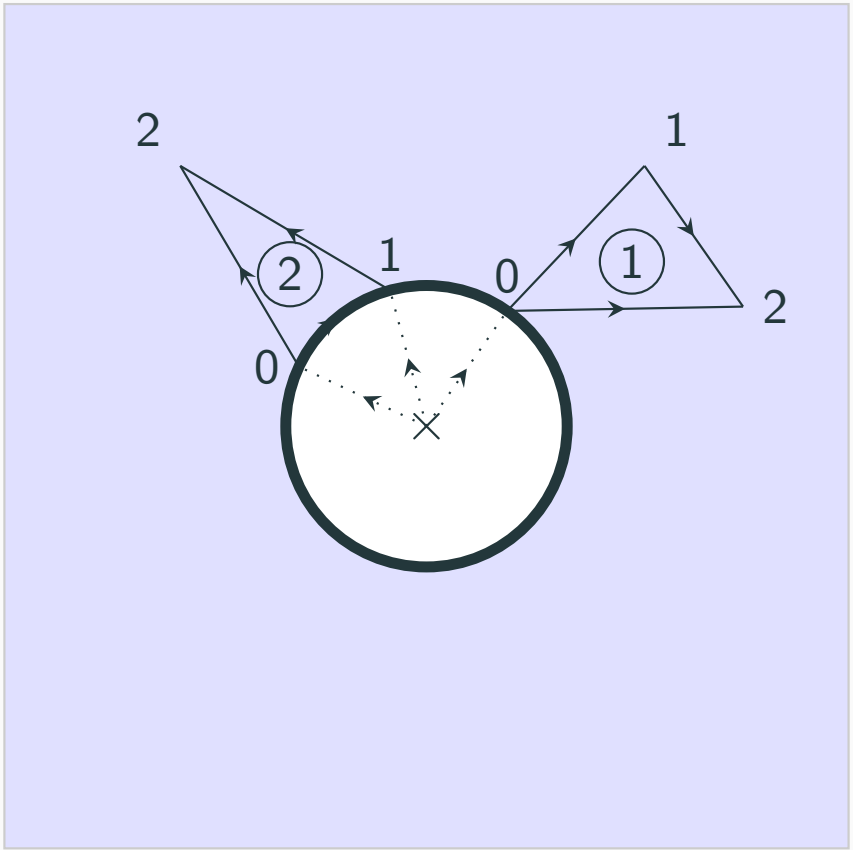
$$\mathfrak{S} = \left(\begin{array}{ccc} & L & \\ {\scriptstyle \pi}\swarrow & & \searrow{\scriptstyle \iota} \\ M & & N \end{array} \right)$$

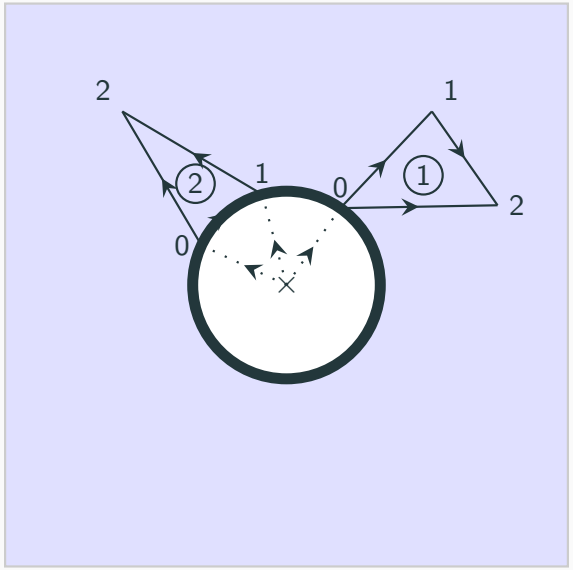
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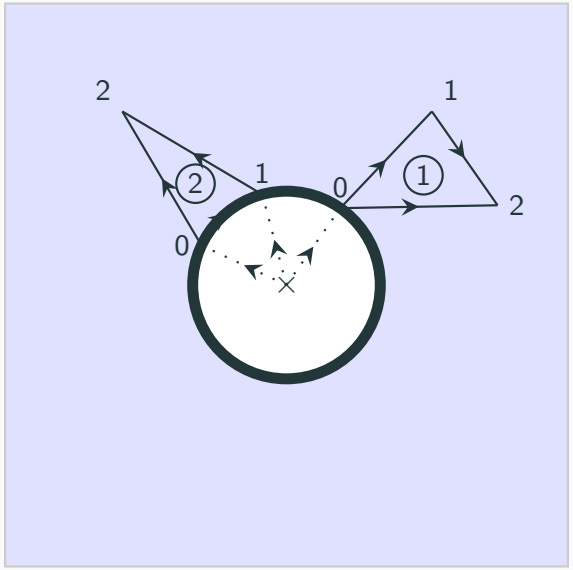
Idea: adjoin non-invertibles formally:





$$\mathrm{EX}(\mathfrak{S}) = M + N + \text{non-invertibles}$$

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$$s_i(\Gamma, j) = (s_i\Gamma, \sharp_{j,i})$$

$$\flat_{j,i} = \begin{cases} j, & i \geq j \\ j-1, & i < j \end{cases} \qquad \sharp_{j,i} = \begin{cases} j, & i \geq j \\ j+1, & i < j \end{cases}$$

Theorem (T '25) $\mathrm{EX}(M \xleftarrow{\pi} L \xrightarrow{\iota} N)$ *is an* ∞*-category whenever* M, L, N *are* ∞*-categories,* π *is a right fibration, and* ι *is a cofibration.*

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Theorem (T '25, Ivanov–T '26) *The image of* EX *includes* $\mathrm{Exit}(M)$ *for conically smooth* M*.*

Over $P = \{0 < 1\}$*:* $\mathrm{Exit}(M) \simeq \mathrm{EX}(M_0 \leftarrow \mathrm{Link} \hookrightarrow \mathrm{Unzip}^\circ \cong M_1)$

The proofs of the counterexamples

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$\mathfrak{S} = (* \leftarrow \Lambda \hookrightarrow \mathbb{R}^N)$ with Λ closed and with $w(\Lambda) \neq 0$

E.g. $\mathbb{R}P^n$ with $n+1 \neq 2^k$ since
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since both Z and Λ are *compact*

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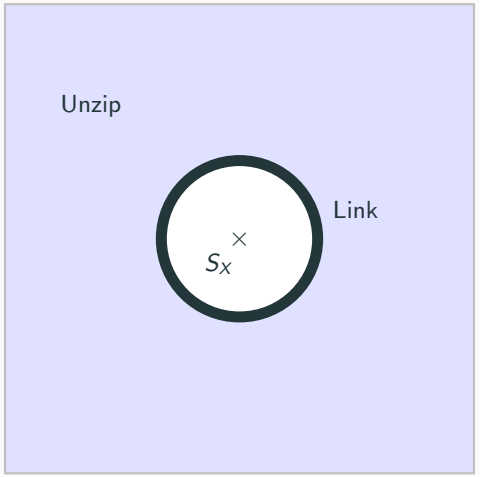
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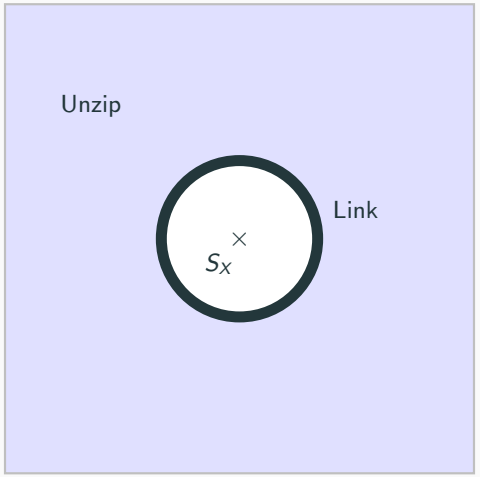
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(Lefschetz duality, using that M is compact; non-balls in 4d: Mazur!)

Outlook: ∞-categories via geometric topology?

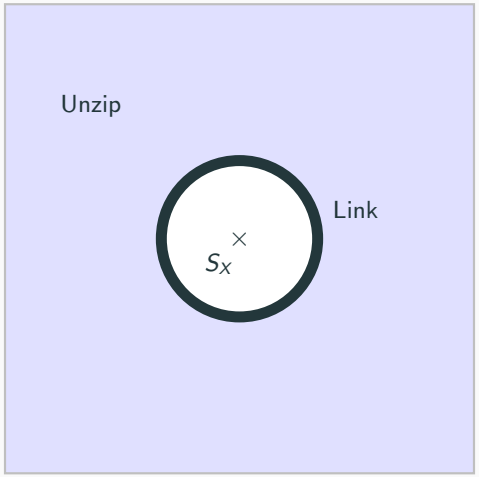


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$M^\circ \cong \mathrm{Unzip}(M)^\circ$ is the interior of a smooth manifold with **corners**.

M° *collapses into* S_M *smoothly at infinity* (but not everywhere)

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No obstruction without the compactness requirement.